

Aditya Kulkarni

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kulkarniaditya.com | Raleigh, NC (local to RTP)

PROFESSIONAL SUMMARY

MS CS student at NC State (May 2026) with hands-on experience in systems programming, device-level C development, and full-stack software engineering. Extended the Xinu OS in C to implement a complete demand-paging virtual memory subsystem -- page fault handler, Second-Chance frame replacement, backing store I/O, TLB management, and 7 new kernel system calls. Complemented by production software engineering at Isomer AI (TypeScript/Node.js, 394 commits, 7 deployments) and distributed systems work with Python, Kafka, and Spark. Local to Research Triangle Park.

CORE COMPETENCIES

C / Systems Programming

OS Internals & Kernel Development

Virtual Memory & Paging

Python (Scripting & Automation)

C++ (OOP & Application Dev)

Java

Full-Stack Engineering (TypeScript/Node.js/React)

Git & CLI Tools

Debugging & Performance Analysis

PROJECTS

Demand Paging Virtual Memory - Xinu OS

Systems Programming · NC State OS Course · C

Extended the **Xinu operating system** in C to implement a full **demand-paging virtual memory subsystem** -- the same architecture underpinning Linux and Windows memory management.

What was built: Each process gets its own isolated virtual address space backed by up to 8 private backing stores (simulated disk). Pages are loaded lazily into physical memory only on access. When RAM is full, a **Second-Chance (Clock) replacement policy** evicts the best candidate frame, writing dirty pages back to the backing store.

Key subsystems implemented: (1) *Page fault handler* -- traps CPU interrupt 14, reads the faulting address from CR2, resolves the backing store via `bsm_lookup()`, and transparently loads the missing page; (2) *Frame allocator* -- manages a circular frame queue with reference bits for Second-Chance eviction; (3) *TLB consistency* -- issues `invlpg` on eviction and reloads CR3 on context switch; (4) *vcreate() / vgetmem()* -- extended process creation and heap allocation APIs for virtual address spaces; (5) *xmmap() / xmunmap()* -- shared memory mapping across processes, analogous to POSIX `mmap`.

Results: 25% reduction in page faults via optimized replacement policy; 15% memory performance improvement; 7 new kernel system calls added to the Xinu syscall table.

C · x86 2-level paging · Page tables · TLB · Demand paging · Second-Chance replacement · Backing store I/O · Kernel syscalls

AlgoPulse - Event-Driven AI Platform

Independent · 2025

Designed and built a distributed AI orchestration platform in TypeScript/Node.js. Identified a polling bottleneck under load and re-architected to event-driven streaming with **Temporal.io** for durable workflow state and **Kafka** for message brokering -- handling 2,000 async AI inference requests/second with zero state loss.

TypeScript · Node.js · React · Temporal.io · Kafka · Supabase Edge Functions

Glance AI - Privacy-First RAG System

Independent · 2025

Built a Python RAG pipeline with per-tenant data isolation at the vector retrieval layer for fintech/healthcare compliance environments. Deterministic retrieval design eliminates cross-tenant context leakage.

Python · LLMs · Vector Databases · RAG

Mapify - ML & NLP Tool

IEEE Published

T5-transformer pipeline that converted 1,500+ PDFs/images into mind maps and flowcharts, reducing academic review time by 60%. Presented to 500+ attendees at IEEE conference.

Python · T5 Transformer · NLTK · NetworkX

EXPERIENCE

Isomer AI

May 2025 - Present

Product Engineer Intern - Remote

- Shipped **10+ production features** across 7 deployments on a TypeScript/Node.js/React/PostgreSQL stack; contributed **394 commits** with high code quality standards and thorough testing
- Built backend pipelines (PDF generation, email parsing, S3 storage), RBAC system, and multi-model AI integrations using clean, maintainable TypeScript with Auth0, Redis, Docker, and Sentry

NC State University

Apr 2025 - Present

Graduate Research Assistant

- Built a cloud-based genomic data platform (React + JBrowse 2) serving 100+ researchers across 500GB datasets on Google Cloud Run; applied ML hyperparameter tuning improving model accuracy by 12%

Mphasis (HP Company)

Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven automation workflows in Python using OCR, NLP, and ML; automated deployments on AWS & GCP with Grafana monitoring dashboards

Capgemini

Jun 2023 - Aug 2023

AI/ML Intern

- Led end-to-end SDLC for a support ticket automation system; built distributed pipelines with **Python, Kafka, and Spark** -- reduced latency by 20% and processing time by 15%

EDUCATION

North Carolina State University - MS Computer Science

Aug 2024 - May 2026

Courses: Artificial Intelligence, **Operating Systems**, Data Science, Software Engineering, Algorithms

Mumbai University - BTech Computer Science

Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: DS & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

Systems / Low-Level: C, C++, x86 paging, kernel development, memory management, TLB, virtual memory, syscalls

Languages: Python, TypeScript, JavaScript, Java, SQL, R

Tools & Workflow: Git, Linux CLI, Docker, CI/CD, Grafana, Sentry, AWS, GCP

Frameworks: Node.js, React, Next.js, Express, PostgreSQL, Redis, Kafka, Spark, Hadoop

AI/ML: PyTorch, TensorFlow, LangChain, OpenAI API, RAG, Hugging Face